

In 1941 the US airline fleets totaled 322 aircraft, mostly DC-3s; over the next four years, more than 11,000 transport versions of the DC-3 were built, and 1,600 transport versions of the DC-4. Although there was an inevitable collapse in production once the war ended, the expanded industrial base was available to supply an expanding commercial market. Equally important was the development of a network of international air routes, with massive investment in infrastructure,

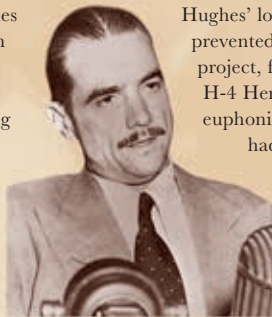
including concrete runways from which four-engined aircraft could safely operate. By the time peace came, the airlines and their personnel had amassed an impressive experience of operating worldwide, including transoceanic services.

During the war a clear vision emerged of a future in which aviation would shrink the globe. For example, American politician Wendell Willkie pronounced: "There are no distant places any longer: the world is small and the world is one."

HOWARD ROBARD HUGHES

WHEN MILLIONAIRE Howard Hughes (1905–76) became a major force in passenger aviation by buying into TWA in 1939, he was already an aviation legend. After first investing his inherited fortune and eccentric talent in Hollywood movies, including the air-combat epic *Hell's Angels*, in 1934 Hughes had sought out the excitements of air racing. He set up Hughes Aircraft to design his own racer, the H-1, in which he established a new world land-plane speed record of 352.3mph (563.7kph) in 1935. Other aviation feats followed, culminating in a record 91-hour around-the-world flight in a Lockheed Super Electra in 1938.

As an airplane manufacturer, though, Hughes was not a success. Convinced that plywood, not aluminum, was the material for the aircraft of the future, he proposed a wooden-framed bomber to the US War Department. They turned it down, as they had the H-1, offered as a fighter.



ECCENTRIC MILLIONAIRE
Howard Hughes was a daring pilot but often a poor judge of aircraft design. In later years his eccentricities got the upper hand over his undoubted talents, and he died a recluse.

Hughes' lobbying strength in Washington prevented immediate refusal of his next project, for a huge military flying boat. The H-4 Hercules, made mainly of birch but euphoniously dubbed the "Spruce Goose," had eight engines and, at 320ft (97.5m), a larger wingspan than any other aircraft before or since. But only a prototype was ever built, and that was two years too late for the war in which it was meant to fight. It rose into the air just once, on November 2, 1947, with Hughes at the controls, reaching an altitude of about 70ft (20m) and covering less than a mile. By that time Hughes' flying career was largely at an end. In 1946 he

was almost killed piloting the XF-11, an all-metal reconnaissance version of his wooden D-2 bomber. After that crash he rarely flew. He lost hands-on control of Hughes Aircraft in 1953 and was elbowed out of the financially ailing TWA in 1960.



SPRUCE GOOSE

Howard Hughes' gigantic H-4 Hercules flying boat, known as the "Spruce Goose," was intended as a military transport aircraft that would carry a large enough cargo in its huge hold to act as a replacement for some of the U-boat-threatened merchant ships plying the Atlantic in World War II.

CRUISING THE STRATOSPHERE

The 377 Stratocruiser was Boeing's last propliner. Its name drew attention to the fact that its pressurized cabin allowed it to cruise in the stratosphere, above the discomfort of the weather. Unfortunately, however, it suffered frequent engine failures.



In 1944, a conference met in Chicago to lay the political and administrative foundations for a new era of international air travel. The US representative, Adolf Berle, saw the Chicago

meeting as a chance to write "the charter of the open sky." This did not quite happen. Countries were not ready to give up control of their airspace or to stop promoting the interests of their national airlines. But enough agreement was reached to let the expansion of international air travel happen.

Increased competition

The international airline industry did not even approximate to a model of free competition. Most airlines either had government subsidy or were state-owned, and the tight regulation of routes and operations was the accepted rule. The

International Air Transport Association (IATA) fixed fares and coordinated the schedules of different companies. The US was the leader in world aviation, and by 1951 more Americans were traveling by air than on long-distance train services. But airlines based in other parts of the developed world held their own. Although Germany was totally excluded from aviation, and

